

Supporting Information for SONIC: Symmetry-Oriented Non-redundant Internal Coordinates

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S1 Standalone Input and Output Reference

S1.1 Input schema `matrix.smith.input.v1`

The standalone boundary is a standard Cartesian XYZ block followed by optional SMITH directives. Four provider-neutral profiles are accepted. Their meaning is fixed by the input contract rather than by the program that created the data.

If `TOPOLOGY` and `PRIMITIVES` are both present, their bond rows must agree. The output records the selected profile, the source fingerprints and the accepted construction options. The minimal route is sufficient for ordinary molecules; interaction centers, externally audited quasi-symmetry or special primitives can be supplied through the richer profiles. This keeps SMITH operationally standalone while making the perception/construction boundary explicit.

Table S1: The four standalone input profiles.

Profile	Required content and action
FROZEN_STATE	A complete validated molecular state. Topology, symmetry operations, atom permutations, primitive rows and provenance are consumed unchanged.
STANDALONE_TOPOLOGY	Cartesian geometry and a supplied TOPOLOGY section. The pinned standalone frontend constructs the ordinary redundant primitives.
STANDALONE_PRIMITIVES	Cartesian geometry and a supplied PRIMITIVES section. Values, atom indices, Wilson rows and geometry fingerprints are validated before SONIC construction.
STANDALONE_MINIMAL	Plain Cartesian geometry. The pinned frontend constructs covalent topology, cycles, point-group operations with atom permutations and ordinary primitives.

S1.2 Norbornane input and Gaussian 16 ReadAllGIC export

The first listing contains only the XYZ geometry and the minimum directives needed for the compatibility export. It is not the enriched **xyzin** container written after construction. The complete workflow is

```
python -m matrix smith standalone norbornane_smith_input.xyzin norbornane.xyzin
python -m matrix smith gaussian-input norbornane.xyzin norbornane_g16_readallgic.gjf \
--title "norbornane SONIC ReadAllGIC optimization" \
--route "#p hf/sto-3g opt=(readallgic,calcf, maxcycle=80)"
```

The reduced standalone distribution provides the equivalent command

```
smith-sonic build norbornane.smith.xyz norbornane.xyzin
```

and writes a human-readable **.smith.out** report together with the Gaussian input.

Listing S1: Minimal norbornane SMITH/SONIC standalone input.

```
19
Norbornane starting geometry for standalone SMITH/SONIC ReadAllGIC export
C 1.25441300 0.78233900 -0.49340600
C 0.00000000 1.13336700 0.33992900
C -0.00000000 -1.13336700 0.33992900
C 1.25441300 -0.78233900 -0.49340600
H 1.20544300 1.20489500 -1.50351600
H 2.16019200 1.17688700 -0.01817100
H 1.20544300 -1.20489500 -1.50351600
H 2.16019200 -1.17688700 -0.01817100
C 0.00000000 0.00000000 1.38887600
H -0.89255700 0.00000000 2.02655300
H 0.89255700 -0.00000000 2.02655300
C -1.25441300 0.78233900 -0.49340600
H -2.16019200 1.17688700 -0.01817100
H -1.20544300 1.20489500 -1.50351600
C -1.25441300 -0.78233900 -0.49340600
H -2.16019200 -1.17688700 -0.01817100
H -1.20544300 -1.20489500 -1.50351600
H 0.00000000 2.15523000 0.73149700
H -0.00000000 -2.15523000 0.73149700

#SMITH
schema = matrix.smith.input.v1
symmetrize = false
```

Listing S2: Complete commercial Gaussian 16-compatible ReadAllGIC input generated from the norbornane contract.

```
#p hf/sto-3g opt=(readallgic,calcf, maxcycle=80) output=pickett

norbornane SONIC ReadAllGIC optimization

O 1
C 1.25441300 0.78233900 -0.49340600
C 0.00000000 1.13336700 0.33992900
C -0.00000000 -1.13336700 0.33992900
C 1.25441300 -0.78233900 -0.49340600
H 1.20544300 1.20489500 -1.50351600
H 2.16019200 1.17688700 -0.01817100
```

H	1.20544300	-1.20489500	-1.50351600
H	2.16019200	-1.17688700	-0.01817100
C	0.00000000	0.00000000	1.38887600
H	-0.89255700	0.00000000	2.02655300
H	0.89255700	-0.00000000	2.02655300
C	-1.25441300	0.78233900	-0.49340600
H	-2.16019200	1.17688700	-0.01817100
H	-1.20544300	1.20489500	-1.50351600
C	-1.25441300	-0.78233900	-0.49340600
H	-2.16019200	-1.17688700	-0.01817100
H	-1.20544300	-1.20489500	-1.50351600
H	0.00000000	2.15523000	0.73149700
H	-0.00000000	-2.15523000	0.73149700

GIC001(Value=0.0) = 0.707106781187*(R(9,10))-0.707106781187*(R(9,11))
 GIC002(Value=2.93156150623) = 0.707106781187*(R(9,10))+0.707106781187*(R(9,11))
 GIC003(Value=1.09633572703) = R(1,6)
 GIC004(Value=1.09633572703) = R(4,8)
 GIC005(Value=1.09633572703) = R(12,13)
 GIC006(Value=1.09633572703) = R(15,16)
 GIC007(Value=1.09602638934) = R(4,7)
 GIC008(Value=1.09602638934) = R(12,14)
 GIC009(Value=1.09602638934) = R(15,17)
 GIC010(Value=1.09602638934) = R(1,5)
 GIC011(Value=1.09431690538) = R(2,18)
 GIC012(Value=1.09431690538) = R(3,19)
 GIC013(Value=1.564678) = R(1,4)
 GIC014(Value=1.564678) = R(12,15)
 GIC015(Value=0.0) = 0.707106781187*(R(3,4))-0.707106781187*(R(3,15))
 GIC016(Value=0.0) = -0.707106781187*(R(1,2))+0.707106781187*(R(2,12))
 GIC017(Value=4.13260235459) = 0.707106781187*(R(3,4))+0.707106781187*(R(3,15))
 GIC018(Value=4.13260235459) = 0.707106781187*(R(1,2))+0.707106781187*(R(2,12))
 GIC019(Value=0.0) = -0.707106781187*(R(2,9))+0.707106781187*(R(3,9))
 GIC020(Value=4.12706024365) = 0.707106781187*(R(2,9))+0.707106781187*(R(3,9))
 GIC021(Value=-0.0573424722848) = 0.816496580928*(A(5,1,6))-0.408248290464*(A(2,1,5))-0.408248290464*(A(2,1,6))
 GIC022(Value=0.0136523036609) = 0.707106781187*(A(2,1,5))-0.707106781187*(A(2,1,6))
 GIC023(Value=-0.0650891780287) = 0.816496580928*(A(5,1,6))-0.408248290464*(A(4,1,5))-0.408248290464*(A(4,1,6))
 GIC024(Value=0.0195508295595) = 0.707106781187*(A(4,1,5))-0.707106781187*(A(4,1,6))
 GIC025(Value=0.0337761539671) = 0.816496580928*(A(18,2,9))-0.408248290464*(A(18,2,12))-0.408248290464*(A(18,2,1))
 GIC026(Value=0.0) = 0.707106781187*(A(18,2,12))-0.707106781187*(A(18,2,1))
 GIC027(Value=0.0337761539671) = 0.816496580928*(A(19,3,9))-0.408248290464*(A(19,3,15))-0.408248290464*(A(19,3,4))
 GIC028(Value=0.0) = 0.707106781187*(A(19,3,15))-0.707106781187*(A(19,3,4))
 GIC029(Value=-0.0650891780287) = 0.816496580928*(A(7,4,8))-0.408248290464*(A(1,4,7))-0.408248290464*(A(1,4,8))
 GIC030(Value=0.0195508295595) = 0.707106781187*(A(1,4,7))-0.707106781187*(A(1,4,8))
 GIC031(Value=-0.0573424722848) = 0.816496580928*(A(7,4,8))-0.408248290464*(A(3,4,7))-0.408248290464*(A(3,4,8))
 GIC032(Value=0.0136523036609) = 0.707106781187*(A(3,4,7))-0.707106781187*(A(3,4,8))
 GIC033(Value=-0.0619062683331) = 0.816496580928*(A(10,9,11))-0.408248290464*(A(2,9,10))-0.408248290464*(A(2,9,11))
 GIC034(Value=0.0) = 0.707106781187*(A(2,9,10))-0.707106781187*(A(2,9,11))
 GIC035(Value=-0.0619062683331) = 0.816496580928*(A(10,9,11))-0.408248290464*(A(3,9,10))-0.408248290464*(A(3,9,11))
 GIC036(Value=0.0) = 0.707106781187*(A(3,9,10))-0.707106781187*(A(3,9,11))
 GIC037(Value=-0.0573424722848) = 0.816496580928*(A(13,12,14))-0.408248290464*(A(2,12,13))-0.408248290464*(A(2,12,14))
 GIC038(Value=-0.0136523036609) = 0.707106781187*(A(2,12,13))-0.707106781187*(A(2,12,14))
 GIC039(Value=-0.0650891780287) = 0.816496580928*(A(13,12,14))-0.408248290464*(A(15,12,13))-0.408248290464*(A(15,12,14))
 GIC040(Value=-0.0195508295595) = 0.707106781187*(A(15,12,13))-0.707106781187*(A(15,12,14))
 GIC041(Value=-0.0573424722848) = 0.816496580928*(A(16,15,17))-0.408248290464*(A(3,15,16))-0.408248290464*(A(3,15,17))
 GIC042(Value=-0.0136523036609) = 0.707106781187*(A(3,15,16))-0.707106781187*(A(3,15,17))
 GIC043(Value=-0.0650891780287) = 0.816496580928*(A(16,15,17))-0.408248290464*(A(12,15,16))-0.408248290464*(A(12,15,17))
 GIC044(Value=-0.0195508295595) = 0.707106781187*(A(12,15,16))-0.707106781187*(A(12,15,17))
 GIC045(Value=-0.0207227541011) = 0.632455532034*(A(3,4,1))-0.511667273602*(A(4,1,2))+0.195439507585*(A(1,2,9))
 +0.195439507585*(A(2,9,3))-0.511667273602*(A(9,3,4))
 GIC046(Value=-0.063778079142) = 0.37174803446*(A(4,1,2))-0.601500955008*(A(1,2,9))+0.601500955008*(A(2,9,3))-0.37174803446*(A(9,3,4))
 GIC047(Value=-0.0207227541011) = 0.632455532034*(A(3,15,12))-0.511667273602*(A(15,12,2))+0.195439507585*(A(12,2,9))
 +0.195439507585*(A(2,9,3))-0.511667273602*(A(9,3,15))
 GIC048(Value=-0.063778079142) = 0.37174803446*(A(15,12,2))-0.601500955008*(A(12,2,9))+0.601500955008*(A(2,9,3))
 -0.37174803446*(A(9,3,15))
 V0001(Value=-55.8772472164) = D(1,2,9,3)
 V0002(Value=55.8772472164) = D(2,9,3,4)
 V0003(Value=-35.2663208258) = D(9,3,4,1)
 V0004(Value=0.0) = D(3,4,1,2)
 V0005(Value=35.2663208258) = D(4,1,2,9)
 V0006(Value=55.8772472164) = D(12,2,9,3)
 V0007(Value=-55.8772472164) = D(2,9,3,15)
 V0008(Value=35.2663208258) = D(9,3,15,12)
 V0009(Value=0.0) = D(3,15,12,2)
 V0010(Value=-35.2663208258) = D(15,12,2,9)

S1.3 Formic-acid–water fragment contract

The eight-atom complex is processed with an automatic connected-component fragment definition and closes at rank 18/18. Six protected rows describe the three relative translations and three relative orientations; the remaining twelve rows span the intramolecular motions. Every build writes the frozen `xyzin`, a readable coordinate report and the optional Gaussian input.

The symmetry-projected fragment path is also checked with exact-family-quota regressions in C_s , C_{2v} , C_{3v} , and C_i . Selection is the intersection of the linear matroid defined by independent analytic Wilson rows and the partition matroid defined by the native SONIC family dimensions. The tests require full rank, unchanged counts for every family, three relative translations and three relative orientations per additional nonlinear fragment, assigned irreps, and successful Cartesian covariance under every stored operation. The C_{2v} , C_{3v} , and C_i probes are algorithmic symmetry fixtures rather than optimized weak-complex structures.

Listing S3: Coordinate-definition excerpt for the formic-acid–water complex.

```
Coordinate Definitions
-----
1 GIC001 Str0001 family=STRETCH class=ORDINARY irrep=A state=ACTIVE value=1.095460024 unit=angstrom components=P001:+1
  Human formula: Str0001 = +1*P001
  +1 * P001: distance R(1,2)
2 GIC002 Str0002 family=STRETCH class=ORDINARY irrep=A state=ACTIVE value=1.211503996 unit=angstrom components=P002:+1
  Human formula: Str0002 = +1*P002
  +1 * P002: distance R(1,3)
3 GIC003 Str0003 family=STRETCH class=ORDINARY irrep=A state=ACTIVE value=1.325944816 unit=angstrom components=P003:+1
  Human formula: Str0003 = +1*P003
  +1 * P003: distance R(1,4)
4 GIC004 Str0004 family=STRETCH class=ORDINARY irrep=A state=ACTIVE value=0.985874998 unit=angstrom components=P004:+1
  Human formula: Str0004 = +1*P004
  +1 * P004: distance R(4,5)
5 GIC005 Str0005 family=STRETCH class=ORDINARY irrep=A state=ACTIVE value=0.9603233822 unit=angstrom components=P005:+1
  Human formula: Str0005 = +1*P005
  +1 * P005: distance R(6,7)
6 GIC006 Str0006 family=STRETCH class=ORDINARY irrep=A state=ACTIVE value=0.9705527871 unit=angstrom components=P006:+1
  Human formula: Str0006 = +1*P006
  +1 * P006: distance R(6,8)
```

Listing S4: Gaussian 16 ReadAllGIC input for formic-acid–water.

```
#p hf/sto-3g opt=(readallgic,calcf, maxcycle=80) output=pickett Pop=Hirshfeld IOp(6/80=1)

formic_acid_water SMITH/SONIC Gaussian 16 optimization

0 1
C -1.17172700 -0.01899900 -0.00137000
H -2.25686900 0.13036900 -0.01510700
O -0.65137600 -1.11304500 0.00496400
O -0.53354400 1.14322800 0.00760300
H 0.43520300 0.96063300 0.01934300
O 1.93014500 -0.02597600 -0.05226900
H 2.59464300 -0.12891700 0.63334500
H 1.35131700 -0.80263400 0.00883100

F001 = Fragment(1-5)
F002 = Fragment(6-8)
CxFO01(Inactive) = XCntr(F001)
CyFO01(Inactive) = YCntr(F001)
CzFO01(Inactive) = ZCntr(F001)
CxFO02(Inactive) = XCntr(F002)
CyFO02(Inactive) = YCntr(F002)
CzFO02(Inactive) = ZCntr(F002)
RPF001(Inactive) = SQRT((X(5)-CxFO01)**2+(Y(5)-CyFO01)**2+(Z(5)-CzFO01)**2)
PxFO01(Inactive) = [X(5)-CxFO01]/RPF001
PyFO01(Inactive) = [Y(5)-CyFO01]/RPF001
PzFO01(Inactive) = [Z(5)-CzFO01]/RPF001
QQxFO01(Inactive) = PyFO01*[Z(3)-CzFO01]-PzFO01*[Y(3)-CyFO01]
QQyFO01(Inactive) = PzFO01*[X(3)-CxFO01]-PxFO01*[Z(3)-CzFO01]
QQzFO01(Inactive) = PxFO01*[Y(3)-CyFO01]-PyFO01*[X(3)-CxFO01]
RQF001(Inactive) = SQRT(QQxFO01**2+QQyFO01**2+QQzFO01**2)
QxFO01(Inactive) = QQxFO01/RQF001
QyFO01(Inactive) = QQyFO01/RQF001
QzFO01(Inactive) = QQzFO01/RQF001
SxFO01(Inactive) = PyFO01*QzFO01-PzFO01*QyFO01
SyFO01(Inactive) = PzFO01*QxFO01-PxFO01*QzFO01
SzFO01(Inactive) = PxFO01*QyFO01-PyFO01*QxFO01
RPF002(Inactive) = SQRT((X(8)-CxFO02)**2+(Y(8)-CyFO02)**2+(Z(8)-CzFO02)**2)
```

```

PxFO02(Inactive) = [X(8)-CxFO02]/RPF002
PyFO02(Inactive) = [Y(8)-CyFO02]/RPF002
PzFO02(Inactive) = [Z(8)-CzFO02]/RPF002
QxFO02(Inactive) = PyFO02*[Z(6)-CzFO02]-PzFO02*[Y(6)-CyFO02]
QyFO02(Inactive) = PzFO02*[X(6)-CxFO02]-PxFO02*[Z(6)-CzFO02]
QzFO02(Inactive) = PxFO02*[Y(6)-CyFO02]-PyFO02*[X(6)-CxFO02]
RQFO02(Inactive) = SQRT(QxFO02**2+QyFO02**2+QzFO02**2)
QxFO02(Inactive) = QxFO02/RQFO02
QyFO02(Inactive) = QyFO02/RQFO02
QzFO02(Inactive) = QzFO02/RQFO02
SxFO02(Inactive) = PyFO02*QzFO02-PzFO02*QyFO02
SyFO02(Inactive) = PzFO02*QxFO02-PxFO02*QzFO02
SzFO02(Inactive) = PxFO02*QyFO02-PyFO02*QxFO02
R11FO02FO01(Inactive) = PxFO02*PxFO01+PyFO02*PyFO01+PzFO02*PzFO01
R12FO02FO01(Inactive) = PxFO02*QxFO01+PyFO02*QyFO01+PzFO02*QzFO01
R13FO02FO01(Inactive) = PxFO02*SxFO01+PyFO02*SyFO01+PzFO02*SzFO01
R21FO02FO01(Inactive) = QxFO02*PxFO01+QyFO02*PyFO01+QzFO02*PzFO01
R22FO02FO01(Inactive) = QxFO02*QxFO01+QyFO02*QyFO01+QzFO02*QzFO01
R23FO02FO01(Inactive) = QxFO02*SxFO01+QyFO02*SyFO01+QzFO02*SzFO01
R31FO02FO01(Inactive) = SxFO02*PxFO01+SyFO02*PyFO01+SzFO02*PzFO01
R32FO02FO01(Inactive) = SxFO02*QxFO01+SyFO02*QyFO01+SzFO02*QzFO01
R33FO02FO01(Inactive) = SxFO02*SxFO01+SyFO02*SyFO01+SzFO02*SzFO01
KwFO02FO01(Inactive) = 0.5*SQRT(R11FO02FO01+R22FO02FO01+R33FO02FO01+1)
KxFO02FO01(Inactive) = (R23FO02FO01-R32FO02FO01)/(4*KwFO02FO01)
KyFO02FO01(Inactive) = (R31FO02FO01-R13FO02FO01)/(4*KwFO02FO01)
KzFO02FO01(Inactive) = (R12FO02FO01-R21FO02FO01)/(4*KwFO02FO01)
KnFO02FO01(Inactive) = SQRT(KxFO02FO01**2+KyFO02FO01**2+KzFO02FO01**2+1.0D-24)
ThFO02FO01(Inactive) = 2*ATAN(KnFO02FO01/KwFO02FO01)
ExFO02FO01(Inactive) = ThFO02FO01*KxFO02FO01/KnFO02FO01
EyFO02FO01(Inactive) = ThFO02FO01*KyFO02FO01/KnFO02FO01
EzFO02FO01(Inactive) = ThFO02FO01*KzFO02FO01/KnFO02FO01
GIC001(Value=1.09546002426) = R(1,2)
GIC002(Value=1.21150399623) = R(1,3)
GIC003(Value=1.32594481625) = R(1,4)
GIC004(Value=0.985874997976) = R(4,5)
GIC005(Value=0.960323382242) = R(6,7)
GIC006(Value=0.9705527871) = R(6,8)
GIC007(Value=123.275171935) = A(2,1,3)
GIC008(Value=110.936908332) = A(2,1,4)
GIC009(Value=179.972129264) = D(2,1,4,3)
GIC010(Value=108.100240696) = A(1,4,5)
GIC011(Value=106.376565278) = A(7,6,8)
GIC012(Value=-179.970146027) = D(2,1,4,5)
GIC013 = CxFO02-CxFO01
GIC014 = CyFO02-CyFO01
GIC015 = CzFO02-CzFO01
GIC016 = ExFO02FO01
GIC017 = EyFO02FO01
GIC018 = EzFO02FO01

```

S1.4 η^3 allyl–palladium interaction-center contract

The ten-atom allyl–PdCl probe consumes a frozen interaction-center record with identifier C001, kind ETA3_CENTER, and atoms 3–5. SMITHretains the Pd–center distance as a protected CENTER_ATOM_DISTANCE; the contract closes at rank 24/24 and the reference value is 1.8000 Å. This is a coordinate-interface test, not an optimized transition-metal benchmark.

Listing S5: Coordinate-definition excerpt for the η^3 allyl–palladium probe.

```

Coordinate Definitions
-----
1 GIC001 CnAt0001 family=CENTER_ATOM_DISTANCE class=SPECIAL_PROTECTED irrep=UNASSIGNED state=ACTIVE value=1.8 unit=
  angstrom components=P012:+1
  Human formula: CnAt0001 = +1*P012
    +1 * P012: center atom distance; function=CENTER_ATOM_DIST atoms=(3,4,5)
2 GIC002 Str0001 family=STRETCH class=ORDINARY irrep=UNASSIGNED state=ACTIVE value=2.3 unit=angstrom components=P001:+1
  Human formula: Str0001 = +1*P001
    +1 * P001: distance R(1,2)
3 GIC003 Str0002 family=STRETCH class=ORDINARY irrep=UNASSIGNED state=ACTIVE value=2.140093456 unit=angstrom components=
  P002:+1
  Human formula: Str0002 = +1*P002
    +1 * P002: distance R(1,3)
4 GIC004 Str0003 family=STRETCH class=ORDINARY irrep=UNASSIGNED state=ACTIVE value=2 unit=angstrom components=P003:+1
  Human formula: Str0003 = +1*P003
    +1 * P003: distance R(1,4)
5 GIC005 Str0004 family=STRETCH class=ORDINARY irrep=UNASSIGNED state=ACTIVE value=2.140093456 unit=angstrom components=
  P004:+1

```

```

Human formula: Str0004 = +1*P004
+1 * P004: distance R(1,5)
6 GIC006 Str0005 family=STRETCH class=ORDINARY irrep=UNASSIGNED state=ACTIVE value=1.334166406 unit=angstrom components=
P005:+1
Human formula: Str0005 = +1*P005
+1 * P005: distance R(3,4)
7 GIC007 Str0006 family=STRETCH class=ORDINARY irrep=UNASSIGNED state=ACTIVE value=1.070046728 unit=angstrom components=
P006:+1
Human formula: Str0006 = +1*P006
+1 * P006: distance R(3,6)
8 GIC008 Str0007 family=STRETCH class=ORDINARY irrep=UNASSIGNED state=ACTIVE value=1.087428159 unit=angstrom components=
P007:+1
Human formula: Str0007 = +1*P007
+1 * P007: distance R(3,7)

```

Listing S6: Gaussian 16 ReadAllGIC input for the η^3 probe.

```

#p hf/sto-3g opt=(readallgic,calcfrc,maxcycle=80) output=pickett Pop=Hirshfeld IOp(6/80=1)

eta3_allyl_palladium SMITH/SONIC Gaussian 16 optimization

O 1
Pd 0.00000000 0.00000000 0.00000000
Cl 0.00000000 0.00000000 -2.30000000
C -1.30000000 0.00000000 1.70000000
C 0.00000000 0.00000000 2.00000000
C 1.30000000 0.00000000 1.70000000
H -1.95000000 0.00000000 2.55000000
H -1.65000000 0.90000000 1.20000000
H 0.00000000 0.00000000 3.09000000
H 1.95000000 0.00000000 2.55000000
H 1.65000000 -0.90000000 1.20000000

Cx001(Inactive) = (X(3)+X(4)+X(5))/3
Cy001(Inactive) = (Y(3)+Y(4)+Y(5))/3
Cz001(Inactive) = (Z(3)+Z(4)+Z(5))/3
GIC001 = SQRT((Cx001-X(1))**2+(Cy001-Y(1))**2+(Cz001-Z(1))**2)
GIC002(Value=2.3) = R(1,2)
GIC003(Value=2.1400934559) = R(1,3)
GIC004(Value=2.0) = R(1,4)
GIC005(Value=2.1400934559) = R(1,5)
GIC006(Value=1.33416640641) = R(3,4)
GIC007(Value=1.07004672795) = R(3,6)
GIC008(Value=1.08742815855) = R(3,7)
GIC009(Value=1.09) = R(4,8)
GIC010(Value=1.07004672795) = R(5,9)
GIC011(Value=1.08742815855) = R(5,10)
GIC012(Value=142.594643369) = A(2,1,3)
GIC013 = L(2,1,4,0,-1)
GIC014 = L(1,3,6,0,-1)
GIC015 = L(1,3,6,0,-2)
GIC016(Value=80.2277614486) = A(1,3,7)
GIC017(Value=114.645858551) = A(4,3,7)
GIC018 = L(1,4,8,0,-1)
GIC019 = L(1,4,8,0,-2)
GIC020 = L(1,5,9,0,-1)
GIC021 = L(1,5,9,0,-2)
GIC022(Value=80.2277614486) = A(1,5,10)
GIC023(Value=114.645858551) = A(4,5,10)
GIC024(Value=0.0) = D(4,3,1,5)

```

The two generated contracts were also read without format translation by the downstream consumers used in the integration tests. Their coordinate realization service constructed direction matrices of dimensions 18×24 and 24×30 , respectively.

S2 Domain-First Special-Coordinate Construction

S2.1 Ownership order and conservative recognition

The construction order tested in this work is

stretches $\longrightarrow$ special-domain inventory $\longrightarrow$ complete special blocks $\longrightarrow$ ordinary residual chart.

The special inventory contains ring and butterfly domains before local XY_n domains, so an atom or primitive already owned by a cyclic source space cannot be reclassified by a center rule. XY_3 requires a four-coordinate acyclic center with three equivalent Y branches and one distinct Z branch. XY_2 requires a three-coordinate center satisfying the analogous (2+1) partition, and the center must not belong to a cycle. Terminal atoms of the same element are exactly permutable. Nonterminal branches must additionally have identical rooted graph environments after removing the center. A fully symmetric XY_4 or XY_3 center has no unique Z and is therefore not relabelled as $XY_3 - Z$ or $XY_2 - Z$, respectively.

S2.2 XY_3 and XY_2 coordinate blocks

For $X(Y_1Y_2Y_3) - Z$, the six primitive valence angles comprise three ZXY_a and three Y_aXY_b rows. SMITH retains a five-dimensional orthonormal local bending block: two E -type combinations of the ZXY_a orbit, the symmetric A_1 -type combination of the Y_aXY_b orbit, and two E -type combinations of that orbit. The missing symmetric angular direction is redundant with the bonded local frame. The torsional coordinate is the common rotation

$$q_{\text{rot}}^{XY_3} = (3N_W)^{-1/2} \sum_{w \in W(Z)} \sum_{a=1}^3 D(w, Z, X, Y_a).$$

Thus the two differential directions present in a three-branch torsional span are carried by the local deformation block rather than duplicated as rotor coordinates.

For an acyclic $X(Y_1Y_2) - Z$ center, the owned angular block contains

$$\frac{2A(Y_1, X, Y_2) - A(Z, X, Y_1) - A(Z, X, Y_2)}{\sqrt{6}}, \quad \frac{A(Z, X, Y_1) - A(Z, X, Y_2)}{\sqrt{2}},$$

together with the oriented out-of-plane row of the same center. Its collective rotor is

$$q_{\text{rot}}^{XY_2} = (2N_W)^{-1/2} \sum_{w \in W(Z)} \sum_{a=1}^2 D(w, Z, X, Y_a).$$

The expected periodicities are three and two for equivalent terminal XY_3 and XY_2 groups. Heterosubstitution, a cyclic XY_2 center, a linear branch or the absence of an admissible W anchor triggers a recorded ONEDIH fallback.

S2.3 Executable interface and regression gates

The optional collective-domain path is requested explicitly:

```
python -m matrix smith gicforge build molecule.xyzin \
--xy3-torsions --xy2-torsions
```

The independent Fortran control uses the corresponding **XY3** and **XY2** keywords. Both implementations store normalized coefficients, domain atoms, anchors, term count, periodicity and fallback reason. Regression tests require identical ordered primitive charts and Wilson rows for the Python/Fortran XY_3 control, a six-term normalized XY_2 rotor for the methylamine control, exclusive ownership of the XY_2 out-of-plane row, and rejection of cyclic, heterosubstituted and non-unique- Z motifs. The established fused, bridged, spiro and cage corpus is rerun to ensure that the new center domains do not alter ring ownership or final vibrational rank.

S3 Gaussian 16 Serialization Details

S3.1 ReadAllGIC route and generated syntax

Gaussian 16 is used as an external interpreter because its GIC syntax accepts general symbolic coordinate expressions. The SONIC blocks examined here are one particular application of that substantially broader syntax; they do not define or delimit the other coordinate models and workflows supported by GIC. All

calculations and parser tests in this Supporting Information use the commercial Gaussian 16 Rev. C.01 installation. The broader GIC functionality reported by Marenich et al. was implemented and tested in a development version of the Gaussian suite; those new capabilities are scientifically interesting but were not available to, and were not used in, the present validation. The compatibility route has the form

```
#p method/basis opt=(readallgic,calcfc,maxcycle=80)
```

CalcFC is optional and was used for the direct coordinate-choice comparison, whereas the three-system solver-consumption table in the main text uses no initial force-constant calculation. The generated block is dependency closed: helper centers and fragment axes precede every expression that uses them, inactive helpers are emitted as frozen functions, and the complete rank-closing coordinate set is retained.

S3.2 Terminal translation of native out-of-plane rows

The native SONIC contract uses the signed curvilinear out-of-plane coordinate

$$U(j; a, b, c) = \arcsin \left[\frac{\hat{r}_{ja} \cdot (\hat{r}_{jb} \times \hat{r}_{jc})}{|\hat{r}_{jb} \times \hat{r}_{jc}|} \right],$$

with the stored atom order fixing the sign. Gaussian 16 Rev. C.01 has no native U primitive, so serialization alone applies $U(I, J, K, L) \mapsto D(J, I, L, K)$. Multi- U ring-puckering combinations remain in the frozen SONIC contract, while the Gaussian file receives their component periodic dihedrals. The translation does not redefine the native coordinate model. When molecular symmetry is retained, all non-totally symmetric SONIC rows are written with **Frozen**; for norbornane this leaves 15 active A_1 rows and 36 frozen rows in the full 51-coordinate contract.

S3.3 Gaussian output excerpt

The complete log is retained in the reproducibility archive. The following excerpt records the optimization progress and normal-termination diagnostics.

Listing S7: Selected Gaussian 16 output lines for the norbornane **ReadAllGIC** run.

```
138: CrdQ2X: Updating Cartesian coordinates based on modified values of the internal ones.
590: SCF Done: E(RHF) = -268.892840294 A.U. after 9 cycles
913: Step number 1 out of a maximum of 80
1272: SCF Done: E(RHF) = -268.893823525 A.U. after 7 cycles
1332: Step number 2 out of a maximum of 80
1425: Optimization completed.
1426: -- Stationary point found.
1853: Normal termination of Gaussian 16 at Wed Jul 8 13:46:24 2026.
```

S4 Local-Pseudosymmetry Extended Validation

The accepted template score is denoted s , its separation from the second-best template by Δs , the acceptance ceiling by $s_{\max} = 0.12$, and the minimum separation by $\Delta s_{\min} = 0.02$. Table S2 contains all 20 ideal/distorted CN 5–9 models. Python and the independent Fortran control path agree on every local group, place the local totally symmetric row first, and close the target rank.

Table S2: Complete high-coordination local-template validation.

CN	Local group	Template	Geometry	s	Δs	$s_{\max}$	$\Delta s_{\min}$	Rank
5	D_{3h}	trigonal bipyramidal	ideal	2.48×10^{-17}	0.217962	0.12	0.02	12
5	D_{3h}	trigonal bipyramidal	distorted	0.007152	0.206418	0.12	0.02	12
5	C_{4v}	square pyramidal	ideal	5.30×10^{-17}	0.217962	0.12	0.02	12
5	C_{4v}	square pyramidal	distorted	0.012670	0.202571	0.12	0.02	12
6	O_h	octahedral	ideal	0	0.375781	0.12	0.02	15
6	O_h	octahedral	distorted	0.006421	0.364851	0.12	0.02	15
6	D_{3h}	trigonal prismatic	ideal	9.72×10^{-17}	0.375781	0.12	0.02	15
6	D_{3h}	trigonal prismatic	distorted	0.005769	0.368504	0.12	0.02	15
7	D_{5h}	pentagonal bipyramidal	ideal	0	0.174704	0.12	0.02	18
7	D_{5h}	pentagonal bipyramidal	distorted	0.005878	0.165714	0.12	0.02	18
7	C_{3v}	capped octahedral	ideal	5.93×10^{-17}	0.174704	0.12	0.02	18
7	C_{3v}	capped octahedral	distorted	0.005464	0.167037	0.12	0.02	18
8	D_{4d}	square antiprismatic	ideal	3.85×10^{-17}	0.296745	0.12	0.02	21
8	D_{4d}	square antiprismatic	distorted	0.004965	0.290042	0.12	0.02	21
8	D_{2d}	dodecahedral-like	ideal	1.33×10^{-16}	0.296745	0.12	0.02	21
8	D_{2d}	dodecahedral-like	distorted	0.008716	0.284759	0.12	0.02	21
9	D_{3h}	tricapped trigonal prismatic	ideal	2.62×10^{-17}	0.112154	0.12	0.02	24
9	D_{3h}	tricapped trigonal prismatic	distorted	0.004833	0.105571	0.12	0.02	24
9	C_{4v}	capped square antiprismatic	ideal	0	0.112154	0.12	0.02	24
9	C_{4v}	capped square antiprismatic	distorted	0.004434	0.106386	0.12	0.02	24

Two named chemical controls use the same frozen decision record. Distorted SF₆ requires deliberately relaxed effective-atomic-number and distance thresholds, demonstrating that pseudosymmetry is an explicit user choice rather than an implicit relabelling.

Table S3: Named high-coordination controls and their declared thresholds.

System	Template	Group	s	Δs	τ_Z	$\tau_r/\text{\AA}$	Rank
Distorted SF ₆	octahedral	O_h	0.006472	0.364380	0.1	0.05	15/15
[ZrF ₈] ⁴⁻	square antiprismatic	D_{4d}	0.000017	0.296729	0.0005	0.001	21/21

The complete record, including CN 10/12 generic fallbacks and the ambiguous near-tie control, is regenerated with

```
export MATRIX_ROOT=/absolute/path/to/MATRIX
python scripts/run_local_salc_validation.py
```

The resulting `data/local_salc_validation.json` records the template scores, margins, thresholds, ranks, rotation/permutation checks and the Python–Fortran row-space residuals.

S5 Gaussian-GIC Motif Tests: Extended Table

Table S4: Complete numerical checks for the Gaussian-GIC-inspired motif tests.

Motif	Generated SONIC object			Numerical contract
Formamide transfer	$R_{\text{NH}}, R_{\text{OH}}, \Sigma r, \Delta r$			For $R_{\text{NH}} = 1.02$ and $R_{\text{OH}} = 1.21$ Å, $\Sigma r = 2.23$ and $\Delta r = -0.19$ Å; sparse rows equal the exact sum/difference.
Li ₂ –Ar scan	Two protected translation axes, polar radius and elliptic quadratic			At $(x, y) = (3, 4)$, $R = 5.000000$, $(x/7.084)^2 + (y/4.718)^2 = 0.898137$, and the elliptic row has six nonzero Cartesian entries.
XY ₃ local block	Projected homogeneous C_{3v} bend block			Rank 3/3; ordered rows are A_1, E, E , with exactly one totally symmetric row.
Weak-complex rotor	Protected fragment translations and orientations			Water-dimer contract rank 12/12; Wilson matrix shape 12×18 ; finite rigid-step residual below 0.02 in the six protected variables.
Ferrocene Cp torsion	Ring-center/Fe	coordinates	over	Rank and analytic- B rank are 57 at all five points; two ring centers persist; sparse density is 0.095–0.101 and the smallest singular value remains above 10^{-8} .
Cyclohexane ring flip	Generated	A_{2g} RPck001	+	Rank 48/48, three puckering rows, three frozen Gaussian expressions, and no hand-written Gcyc, Ccyc or Scn0.
Reduced path subgroup	E_u RPck001/002 block Same cyclohexane source rebuilt with retained group C_1			Rank remains 48/48; global high-symmetry labels are not imposed along the path and scan torsions remain uncombined.

These checks concern automatic construction and analytic rows, not relative optimizer performance.

S6 Cross-Code Verification (ORCA/Molpro)

The cyclohexane chair-to-boat fixture was exported as ordinary Cartesian geometry and evaluated independently with ORCA and Molpro at PBE0/def2-SVP. The three points are the same fixed-frame interpolation used in the main-text figure: $p00$, $p02$, and $p04$ correspond to $\lambda = 0.00, 0.50, 1.00$. Absolute energies differ because the programs have independent implementations and defaults, but their relative profiles agree within 0.13 kJ mol⁻¹.

Table S5: PBE0/def2-SVP single-point energies on SONIC-generated cyclohexane geometries.

Point	ORCA E_h	ORCA ΔE	Molpro E_h	Molpro ΔE
$p00$	-235.422463982	0.000	-235.422142275	0.000
$p02$	-235.321345943	265.485	-235.321038853	265.447
$p04$	-235.103704314	836.903	-235.103335256	837.028

S7 Pagodane, Indane and Saccharin Ring-Coordinate Fixtures

The complete internal validation corpus underlying the representative main-text table is collected in Table S6. Horizontal rules separate the individual systems so that rank closure, coordinate family and numerical audit can be followed independently.

Table S6: Complete internal validation of the SONIC construction. Symbols and numerical diagnostics are defined in the main text.

System	Stress test	PG	Rank/closure	Numerical check
Benzene	Aromatic ring projector	D_{6h}	30/30; 3 <i>RPck</i>	$\Delta_{PF} = 6.67 \times 10^{-9}$; $\epsilon_{SALC} = 2.22 \times 10^{-16}$
Pyrrole	Heteroaromatic ring projector	C_{2v}	24/24; 2 <i>RPck</i>	$\Delta_{PF} = 6.92 \times 10^{-9}$; $\epsilon_{SALC} = 2.22 \times 10^{-16}$
Azulene	Non-benzenoid fused ring, butterfly	C_{2v}	48/48; 6 <i>RPck</i>	$\Delta_{PF} = 6.44 \times 10^{-9}$; $\epsilon_{SALC} = 2.22 \times 10^{-16}$
Norbornane	Bridged saturated ring	C_{2v}	51/51; 3 <i>RPck</i>	$\Delta_{PF} = 6.22 \times 10^{-9}$; $\epsilon_{SALC} = 2.22 \times 10^{-16}$
Cubane	High-symmetry cage	O_h	42/42; cyclic bends	$\Delta_{PF} = 6.41 \times 10^{-9}$; $\epsilon_{SALC} = 2.22 \times 10^{-16}$
Ferrocene	D_{5h} metallocene projector	D_{5h}	57/57	$\Delta_{PF} = 6.71 \times 10^{-9}$; $\epsilon_{SALC} = 3.33 \times 10^{-16}$
Spiro system	Spiro ring source spaces	D_2	87/87; 6 <i>RPck</i>	$\Delta_{PF} = 6.97 \times 10^{-9}$; $\epsilon_{SALC} = 2.22 \times 10^{-16}$
Cyclooctane	Large-ring puckering	D_2	66/66; 5 <i>RPck</i>	$\Delta_{PF} = 9.84 \times 10^{-9}$; $\epsilon_{SALC} = 2.22 \times 10^{-16}$
Pagodane	Cage and fused-ring source spaces	D_{2h}	114/114	$\epsilon_{FD} = 4.02 \times 10^{-10}$; stretches and cyclic bends covered
Indane	Fused aromatic/alicyclic rings	C_s	51/51	$\epsilon_{FD} = 4.77 \times 10^{-10}$; butterfly and <i>RPck</i> rows covered
Saccharin	Planar fused heterocycle, butterfly-puckering phase	C_s	45/45; 5 <i>RPck</i>	$\epsilon_{FD} = 3.91 \times 10^{-10}$; full post-projector B rank 45
Formic acid–water	Weak-complex fragment model	C_1	18/18; 6 protected	three translations plus three orientations; direct B -rank = 18
Stacked water fragments	Intermolecular symmetry projector	C_{2v}	12/12; 6 protected	native family quotas and Cartesian irrep covariance retained
Separated ammonia–borane fragments	Intermolecular symmetry projector	C_{3v}	18/18; 6 protected	degenerate <i>E</i> subspace, native family quotas and full rank retained
Inversion-related ammonia fragments	Intermolecular symmetry projector	C_i	18/18; 6 protected	$9A_g + 9A_u$; native family quotas and full rank retained
Water dimer	Hydrogen-bonded fragment model	C_1	12/12; 6 protected	standalone Cartesian input; three translations plus three orientations
Benzene–water	Aromatic–polar fragment model	C_1	39/39; 6 protected	standalone Cartesian input; ring coordinates coexist with the fragment block
η^3 allyl–PdCl	Supplied η^3 ligand center	C_1	24/24; 1 protected	$q_{\eta^3} = 1.8000$ Å; analytic center–atom row retained
Weak-complex pseudo-cycles	Intermolecular and intramolecular pseudo-bond contacts	C_s , C_{3v} , C_1	rank closed	formic acid–water, water dimer, $\text{Li}_2\text{–Ar}$, borane–ammonia and glycine I/II fixtures validate the current baseline
Semantic layer	PROTECT/OBSERVABLE contract	–	unit probes	duplicate protected row rejected; row-equivalent substitution recorded; protected-rank excess fails fast

Three Cartesian structures are retained verbatim as publication fixtures. The first is pagodane, $\text{C}_{20}\text{H}_{20}$: the pinned standalone frontend finds 50 covalent bonds, 11 rings, no aromatic atoms and a D_{2h} reference assignment. The redundant primitive contract and SONIC basis both have Cartesian rank 114. The second is indane, C_9H_{10} : the frontend finds 20 bonds, two fused rings, six aromatic atoms and a C_s reference assignment, while SONIC closes at rank 51.

The third fixture is planar saccharin, retained specifically as a fused-ring phase and global-rank regression. Its C_s contract contains one butterfly and five ring-puckering coordinates, all in A'' , and closes at rank 45/45. The generated butterfly is

$$AppBtFl001 = 2^{-1/2}[D(6, 1, 2, 13) - D(16, 1, 2, 3)], \quad (1)$$

whose relative minus sign supplies the inter-ring direction missing from the five *RPck* rows. Replacing it by the corresponding sum lowers the rank of these six rows from six to five. The regression therefore checks the complete post-symmetry Cartesian **B**, rather than accepting six individually well-formed but collectively dependent expressions.

Central finite differences of every final SONIC row give maximum absolute analytic-row errors of 4.02×10^{-10} for pagodane and 4.77×10^{-10} for indane, and 3.91×10^{-10} for saccharin. The indane and saccharin sets contain stretch, bend, cyclic-bend, butterfly and ring-puckering-component families. All 114 pagodane and 51 indane display motions are constructed in both signs with the

Wilson matrix frozen at the reference geometry. They are first-order visualization artifacts rather than nonlinear internal-to-Cartesian realization services.

Extended independent use by collaborators exposed an important earlier failure mode. A dihedral-based representation of endocyclic out-of-plane deformation did not converge in the back-transformation of a near-planar fused indane/indene-type scaffold containing one sp^3 CH₂ center. The ill-conditioning belonged to the collectively redundant cyclic-dihedral chart at planarity, not to SONIC rank or symmetry construction. A native signed U source has a finite first-order row at the planar reference and avoids the dihedral branch ambiguity. Symmetry projection acts on this oriented row space and retains both source and target phases; the complete $\mathbf{B}$ rank is then checked across all coordinate families. The indane and saccharin fixtures are therefore complementary regressions for the change prompted by independent use.

The version-2 diagnostic manifest records the right-inverse metric, mass provenance, coordinate conditioning, local domain and pseudogroup, independent local/global total-symmetry labels, canonical ring order and primitive coefficients. A distributed JSON Schema and dependency-free validator define the machine contract, and the reader remains compatible with archived version-1 manifests.

S8 Reproducibility Details

The standalone manuscript snapshot is tag v0.1.0rc7. Its scientific implementation provenance points to clean MATRIX commit bc6d62140aab, which contains the oriented native-out-of-plane projector and global post-projection rank audit used for the saccharin regression. The declared standalone requirements are Python ≥ 3.11 , NumPy ≥ 1.26 , and setuptools ≥ 68 ; the validation environment used Python 3.13.11, NumPy 2.4.2 and setuptools 80.9.0.

```
python -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install ./standalone
python -m pytest standalone/tests
smith-sonic build standalone/examples/norbornane.smith.xyz norbornane.xyzin

export MATRIX_ROOT=/absolute/path/to/MATRIX
python scripts/run_local_salc_validation.py

latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error supporting_information.tex
```

Representative SHA-256 fingerprints are:

Table S7: Checksums of the standalone boundary and principal validation artifacts.

Artifact	SHA-256
standalone/pyproject.toml	beca83a4eeacd91995b6cedc41fb6af14f1e378c785119b5a5af544c6b2a5f25
data/local_salc_validation.json	aa3208ae1c904f476968d1df00312d1c5c5cbf93caecc4fc1269b4762bb691f9
si/norbornane_smith_input.xyzin	e5feb34cf4c2b1114b65456704342e0c22b687fce85bdb7d4c1e45bbd05885c6
si/norbornane_g16_readallgic.gjf	140742865ae2662c5f74099d78d099e83ab3b65c2a27c75b5d1c4a15996e3857
si/norbornane_g16_output_excerpt.txt	93e21fe46c4d197079221fc5875d1994e5cfcadad2f0c252dd7789709107c4f5

No command or input file requires a path from the author’s workstation; users select the MATRIX checkout, environment and output directories explicitly.